

Carlos De La Cruz

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SUMMARY OF QUALIFICATIONS

Software Engineer building backend systems, applied AI workflows, and agent-runtime infrastructure. Led development of a ML pipeline that boosted model accuracy from 0.43 to 0.92 and cut lab testing time by ~30%, and architected the core layout system for an educational platform serving 100,000+ students. Seeking to apply a blend of applied AI and full-stack expertise to drive data-driven product innovation.

WORK EXPERIENCE

TI Automotive

Nov 2025 - May 2026

Applied AI Engineer / Student Researcher

Toronto, Ontario, Canada

- Built a machine-learning pipeline for polymer-blend recommendation using Python, PyTorch, and model-based reinforcement learning, which accurately predicted material properties from OEM target specifications
- Designed physics-informed feature engineering that improved model performance R^2 from 0.43 to 0.92.
- Enhanced sparse industrial trial data, turning ~470 trial records into 4,136 row training set for experimentation.
- Collaborated with subject matter experts to encode plant/process knowledge into features and reduce lab testing time by ~30%.

Xello

Jun 2025 - May 2026

Software Engineer

Toronto, Ontario, Canada

- Architected the core layout system for The Real Game, organizing activities, containers, and reusable UI structure serving 100,000+ students.
- Shipped TypeScript backend endpoints, SQL operations, Jenkins deployments, and Jasmine tests in a 20-engineer product team.
- Diagnosed and documented a DNS/proxy issue blocking BrowserStack across the team's VM environment, unblocking 50+ engineers.

Exactera

Nov 2022 - Jan 2024

Software Development Engineer II

New York, USA

- Developed the CompSearch Engine using OpenSearch vector search to locate, rank, and group millions of companies by industry, financial forecasts, and comparable attributes, improving comparables search by ~35%.
- Enhanced the UI and API of the Transfer Pricing Platform to speed up report generation using Node.js, Express.js, SQL, and React.js.
- Maintained the company's cloud infrastructure by writing reusable Terraform modules for AWS Lambda and Fargate, enabling rapid spin-up of cloud environments and reducing deployment time by 8%.

SELECT PROJECTS

TryAgent - Multi-tenant Agent Runtime Platform tryagent.ca

- Built a multi-tenant control plane for isolated agent boxes using TypeScript, Effect TS, Go, and OVHcloud.
- Operates five small runtime nodes and generates roughly \$120 MRR.

Ntornos [Blogpost](#)

- Developed a real estate platform with a custom web API microservice architecture using PlanetScale, Prisma, and tRPC, along with an interactive map-based UI featuring Next.js and TailwindCSS.
- Delivered a full-stack property management dashboard enabling fast queries for users and realtors.

EDUCATION

Centennial College

Sep 2023 - May 2026

Advanced Diploma, Software Engineering Technology – AI/ML

Toronto, ON

- **GPA:** 4.4/4.5 (High Honours)
- **Coursework:** Deep Learning, Supervised Learning, Data Structures and Algorithms, Advanced Database Concepts (Oracle SQL), Reinforcement Learning, Linear Algebra & Statistics

Pontificia Universidad Católica Madre y Maestra

Jan 2014 - Dec 2019

Bachelor of Business Administration, Business Administration

Santo Domingo, DR

- **GPA:** 3.5

Fullstack Academy

Jan 2020 - Jun 2020

Software Engineering Immersive

New York

- **Coursework:** Postgres, TypeScript, Express.js, Node.js, React.js